

Thura Win Kyaw

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Profile

I'm an undergraduate researcher working on multimodal AI and small Vision-Language Models (VLMs), with a focus on reasoning faithfulness, grounding, and reliability. My work explores reinforcement learning and preference-based post-training, including GRPO and DPO, with an emphasis on data-, label-, and memory-efficient methods. I'm also interested in 3D vision, computational imaging, and scientific AI. I'm completing a B.E. in Electrical Engineering at Chungnam National University and work as an AI/ML Engineer Intern at AidanBio, building LLM/RAG systems, AI backends, and biomedical AI applications.

Research Interests

Multimodal AI; Vision-Language Models; Efficient Post-Training; Grounded Reasoning; Computational Imaging; 3D Vision.

Research Projects

AidanMED – Label-Efficient VLM Post-Training for Faithful, Grounded Medical Reasoning

- Built during my internship at AidanBio, this project trains a VLM with GRPO to produce reasoning together with a bounding box that must remain faithful to the model's own reasoning.
- Status: In progress. SFT cold-start complete; GRPO post-training is running on 8× NVIDIA A100 GPUs. No evaluation results reported yet.

ChatPDB – Protein Structure Visualization Assistant (LLM + RAG + PyMOL)

- Built an assistant at AidanBio that interprets natural language queries to visualize protein structures with LLM + RAG and PyMOL rendering.
- Implemented a retriever pipeline (BM25 + FAISS + Cross-Encoder) using UniProt annotations and enabled session-based workflows.
- Integrated FastAPI services and a CLI using local LLMs (Ollama) for interactive usage.

PhysHolo – Physics-Informed Holographic Phase Reconstruction

- Explores synthetic hologram-to-phase reconstruction by comparing classical back-propagation, a compact U-Net, and U-Net training with a physics-consistency loss through differentiable Angular Spectrum propagation.
- Status: In progress. Current focus: simulator validation and reconstruction baselines. Evaluation targets include reconstruction accuracy, hologram consistency, noise robustness, and unseen propagation distances.

MoonDPO – Memory-Efficient DPO for Hallucination Mitigation in Small VLMs 

- Applies LoRA-based DPO to Moondream2 (1.9B) using preference pairs automatically generated from COCO annotations, without human or API labels.
- Derived and implemented the DPO objective from scratch; the full pipeline runs on a single consumer GPU (16–24 GB). Trained on 1,945 preference pairs from 5,000 images with LoRA $r=32$ for 3 epochs.
- Preliminary results (single seed; 500 POPE questions per split; 200 CHAIR images): POPE F1 improves across all three splits (Popular 0.900 $\rightarrow$ 0.915), with recall approximately maintained (0.932 $\rightarrow$ 0.928); CHAIR_i 0.132 $\rightarrow$ 0.031, CHAIR_s 0.215 $\rightarrow$ 0.035.
- The CHAIR reduction accompanies fewer objects per caption (1.86 $\rightarrow$ 1.12), indicating a caption-richness trade-off. Training/evaluation image-overlap checks, multi-seed evaluation, and a VQA regression check remain pending.

Experience

AI/ML Engineer Intern, [AIdanBio](#) – Daejeon, KR

Dec 2024 – Present

- Developed AI backend systems, APIs, model-serving infrastructure, and LLM-powered applications.
- Built RAG pipelines using vector databases, semantic search, prompt engineering, and domain-specific knowledge bases.
- Fine-tuned Vision-Language Models (VLMs) for medical AI using multimodal SFT, grounding, GRPO, and evaluation pipelines.
- Worked on knowledge distillation and teacher–student training to transfer reasoning from large multimodal models to smaller models.

AI Software Engineer Intern, [GRINDA AI](#) – Daejeon, KR

Oct 2025 – Nov 2025

- Built an AI-powered Slack bot that converts issue reports into structured GitHub issues using Claude AI and FastAPI, with auto-labeling, translation, and monitoring

Education

[Chungnam National University](#), B.E. in Electrical Engineering

Sept 2022 – Present

- Relevant Coursework: Sensors and Measurement, Electromagnetics, Electronic Circuits, Linear Algebra, Computer Programming, Artificial Intelligence and Data Utility

Skills

ML / DL: PyTorch, Transformers, PEFT/LoRA, NumPy, Scikit-learn.

VLM / Multimodal AI: Vision-Language Models, Visual Grounding, Hallucination Mitigation, Multimodal Reasoning.

Post-Training: SFT, GRPO, DPO, LoRA, Preference Data Generation, Knowledge Distillation.

Compute / Infrastructure: Multi-GPU Training, Linux, Git, GitHub.

Applied AI Systems: RAG, Vector Databases, FastAPI, Model Serving.

Programming Languages: Python, C, MATLAB.

Engineering Simulation: Ansys.

Languages: Burmese (Native), English (Fluent), Korean (TOPIK 5).

Foundational ML

- Iris Flower Classification (SVM) 
- Customer Segmentation (K-means) 
- Text Classification 
- Quote Sentiment Analysis 

References

Available upon request.